

# The Radial-Cyclic Field Model (RCFM)

A 4-Dimensional Topological Framework  
for Cyclic Cosmology

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## Abstract

The Radial-Cyclic Field Model (RCFM) proposes a cosmology in which the universe is a 4-dimensional hyperspherical standing wave. Matter is generated at a continuously active central singularity, streams outward as field-bonded matter on expanding 3-spherical shells (Phase A), undergoes a phase transition at a critical radius  $R_{\max}$  where quantum fields can no longer be sustained, and returns inward as non-interacting ghost quanta (Phase B).

Starting from a metric ansatz on the 4-ball  $B^4$ , we derive a closed system of modified Friedmann equations incorporating a dual-stream energy-momentum tensor with a gravitational drag interaction. We show that:

- (i) Newtonian gravity is recovered exactly in the weak-field, sub-horizon limit.
- (ii) The Phase B ghost quanta naturally produce an effective equation of state  $w_B \approx -1$ , mimicking a small positive cosmological constant without fine-tuning.
- (iii) The singularity boundary condition  $a(r) \propto r^2$  is self-consistent and generates a nearly scale-invariant primordial power spectrum with spectral tilt  $n_s - 1 = -\epsilon_1/k_*$ , making inflation unnecessary.
- (iv) The drag kernel  $\Gamma_{\text{drag}}$  is spatially constant, yielding a fully closed ODE system, and is shown to be perturbatively stable to first order in deviations from the exact  $a \propto r^2$  solution.
- (v) Gravitational wave propagation is consistent with the GW170817 speed constraint  $|c_T - c|/c < 5 \times 10^{-16}$  to order  $\Lambda_{\text{RSD}} < 1.7 \times 10^{-16}$ ; this bound is shown to be naturally satisfied without fine-tuning for Planck-suppressed drag coupling.
- (vi) Thermodynamic entropy is genuinely reset to zero at each passage through the singularity via a Bogoliubov transformation on the boundary field theory; global entropy conservation is maintained through entanglement transfer, in analogy with Page's theorem.

- (vii) The critical transition density  $\rho_c$  is derived from electroweak symmetry restoration via a Ginzburg–Landau effective potential, removing it as a free parameter.
- (viii) Phase B ghost quanta are identified as decoherent momentum eigenstates arising when the gauge condensate vanishes, with their pressurelessness and non-interaction following from the absence of gauge dressing.
- (ix) The spectral tilt parameter  $\epsilon_1$  is derived from first principles via the RCFM Friedmann equation, yielding a joint constraint on  $(\rho_{B,0}, R_{\max}, a_1)$  consistent with the Planck 2018 measurement  $n_s = 0.9649 \pm 0.0042$ .

Quantitative predictions for the CMB angular power spectrum  $C_\ell$ , the matter power spectrum  $P(k)$ , and the stochastic gravitational wave background are derived analytically. Numerical integration of the background equations and Boltzmann solver implementation are identified as the remaining steps before comparison with Planck, DESI, and LIGO/Virgo observational data.

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# 1 Introduction

The standard  $\Lambda$ CDM model provides an extraordinarily successful description of the observable universe, yet leaves several fundamental questions unanswered: the physical origin of the cosmological constant, the mechanism driving inflation, the nature of the thermodynamic arrow of time, and the ultimate fate of the universe [7, 14, 17].

Cyclic cosmologies offer an alternative framework in which the universe undergoes periodic contractions and expansions, potentially resolving the entropy paradox and the flatness problem without appealing to a single fine-tuned initial condition [2, 12, 15].

The Radial-Cyclic Field Model (RCFM) proposes a specific realisation of cyclic cosmology in which the universe is topologically a closed 4-dimensional ball  $B^4$ . The radial coordinate  $r \in [0, R_{\max}]$  serves as an emergent time parameter for observers in the outward-streaming matter phase. The model introduces two coexisting phases at every radial coordinate:

**Phase A** Active matter — all Standard Model gauge fields active, streaming outward.

**Phase B** Ghost quanta — gauge fields dissolved, pressureless, streaming inward.

The interaction between these streams, mediated purely gravitationally, is hypothesised to produce both attractive gravity and cosmic acceleration as two aspects of a single radial drag mechanism.

This paper presents the complete mathematical framework of the RCFM, deriving all equations from first principles and identifying the unique observational signatures that distinguish the model from  $\Lambda$ CDM. A series of previously open theoretical problems—the microphysical origin of the phase transition, the nature of ghost quanta, the perturbative robustness of the drag rate, the thermodynamic consistency of entropy resetting, the naturalness of the GW170817 bound, and the first-principles derivation of the spectral tilt—are each resolved analytically within the framework.

**Organisation.** Section 2 establishes the foundational topology and metric, including the Ginzburg–Landau derivation of the critical density. Section 3 constructs the dual-stream energy-momentum tensor and identifies the microphysical origin of Phase B. Section 4 derives the RCFM field equations. Section 5 specifies the Phase A equation of state. Section 6 proves the weak-field limit. Section 7 derives the CMB angular power spectrum. Section 8 derives the matter power spectrum. Section 9 treats the singularity as a quantum boundary, derives the primordial spectrum, and obtains  $\epsilon_1$  from first principles. Section 10 derives the drag kernel from microphysics and proves its perturbative stability. Section 11 resolves the entropy paradox and addresses information conservation. Section 12 treats gravitational wave propagation. Section 13 collects observational predictions and addresses naturalness. Section 14 outlines the numerical integration programme. Section 15 concludes.

## 2 Foundational Topology and Metric

### 2.1 The 4-Ball Manifold

The universe is modelled as a Lorentzian 4-manifold  $(\mathcal{M}, g)$  where  $\mathcal{M} \cong B^4$  is the closed 4-dimensional ball with radial coordinate  $r \in [0, R_{\max}]$ . The spatial sections at

fixed  $r$  are 3-spheres  $S^3$ .

The metric ansatz is:

$$ds^2 = -A(r) dr^2 + a(r)^2 d\Omega_3^2 \quad (1)$$

where  $a(r)^2 \equiv B(r)$  is the scale factor,  $A(r)$  is the lapse function, and

$$d\Omega_3^2 = d\chi^2 + \sin^2\chi (d\theta^2 + \sin^2\theta d\phi^2) \quad (2)$$

is the round metric on the unit  $S^3$ . The coordinates  $(\chi, \theta, \phi)$  parametrise the angular directions.

This metric is formally identical to a closed Friedmann–Lemaître–Robertson–Walker (FLRW) metric with  $k = +1$  and lapse  $N(r) = \sqrt{A(r)}$ , with  $r$  playing the role of cosmic time.

## 2.2 Radial Time and Proper Time

For comoving observers in Phase A (outward stream), the radial coordinate  $r$  is the emergent time. Proper time along a comoving worldline is:

$$d\tau = \sqrt{A(r)} dr \quad (3)$$

The generalised Hubble parameter is:

$$\mathcal{H}(r) \equiv \frac{1}{\sqrt{A}} \frac{da}{dr} = \frac{a'}{\sqrt{A}} \quad (4)$$

where primes denote  $d/dr$ . In synchronous gauge  $A(r) = 1$ , this reduces to  $\mathcal{H} = a'$ .

## 2.3 The Singularity at $r = 0$

The centre  $r = 0$  is not a past event but a *continuously active boundary condition* — a locus at which infalling Phase B quanta are reprocessed into outgoing Phase A matter. This is analogous to a standing wave node: energy passes through it cyclically while the node itself is stationary in the radial coordinate.

The critical physical consequence is that the singularity acts as a *quantum reflecting boundary*, whose scattering matrix determines the primordial power spectrum (Section 9).

## 2.4 The Critical Radius $R_{\max}$ and the Phase Transition

At  $r = R_{\max}$ , the energy density of the expanding shell falls below the threshold  $\rho_c$  required to sustain gauge field coherence:

$$\rho_A(R_{\max}) = \rho_c \quad (5)$$

The volume of a 3-sphere of radius  $r$  is  $V_3(r) = 2\pi^2 r^3/3$ , so for a shell of fixed energy  $E_{\text{shell}}$ :

$$\rho_A(r) = \frac{3E_{\text{shell}}}{2\pi^2 r^3} \propto r^{-3} \quad (6)$$

The condition (5) therefore defines  $R_{\max}$  uniquely given  $E_{\text{shell}}$  and  $\rho_c$ .

## 2.5 Derivation of $\rho_c$ from Electroweak Physics

In the original formulation,  $\rho_c$  was postulated. We now derive it from first principles using a Ginzburg–Landau effective field theory for the gauge sector order parameter [6, 9].

Let  $\phi$  denote the gauge-sector vacuum expectation value (VEV). The density-dependent effective potential is:

$$V(\phi, \rho_A) = -\frac{m_0^2}{2} \left(1 - \frac{\rho_c}{\rho_A}\right) \phi^2 + \frac{\lambda}{4} \phi^4 \quad (7)$$

This yields two distinct phases depending on the sign of the mass term:

**Phase A** ( $\rho_A > \rho_c$ ): The squared-mass coefficient is positive, so the VEV is non-zero:

$$\langle \phi \rangle = m_0 \sqrt{\frac{1 - \rho_c/\rho_A}{\lambda}} \neq 0 \quad (8)$$

Gauge bosons acquire mass via the Higgs mechanism; all Standard Model interactions are active.

**Phase B** ( $\rho_A < \rho_c$ ): The mass term flips sign;  $\langle \phi \rangle = 0$  is the global minimum. Spontaneous symmetry is *restored*; gauge bosons become massless and decohere. Standard Model interactions dissolve.

The transition is second-order because  $\langle \phi \rangle$  vanishes continuously as  $\rho_A \rightarrow \rho_c^+$ , consistent with the paper’s original claim.

The critical density is fixed to the electroweak scale:

$$\rho_c \equiv \rho_{\text{EW}} = \frac{\pi^2}{30} g_*(T_{\text{EW}}) T_{\text{EW}}^4 \approx \frac{\pi^2}{30} \times 106.75 \times (100 \text{ GeV})^4 \quad (9)$$

This is not a free parameter; it is determined entirely by the measured electroweak scale and the Standard Model effective degrees of freedom. Consequently,  $R_{\text{max}}$  is also fully determined:

$$R_{\text{max}} = \left( \frac{3E_{\text{shell}}}{2\pi^2 \rho_{\text{EW}}} \right)^{1/3} \quad (10)$$

removing one free parameter from the model.

## 3 The Dual-Stream Energy-Momentum Tensor

### 3.1 4-Velocities

Phase A matter streams outward; Phase B ghost quanta stream inward. Their 4-velocities, normalised with  $g_{\mu\nu} u^\mu u^\nu = -1$ , are:

$$u^{(A)\mu} = \left( \frac{1}{\sqrt{A}}, 0, 0, 0 \right), \quad u_\mu^{(A)} = \left( -\sqrt{A}, 0, 0, 0 \right) \quad (11)$$

$$u^{(B)\mu} = \left( -\frac{1}{\sqrt{A}}, 0, 0, 0 \right), \quad u_\mu^{(B)} = \left( +\sqrt{A}, 0, 0, 0 \right) \quad (12)$$

These are antiparallel in the radial direction.

### 3.2 Individual Phase Contributions

The stress-energy of Phase A (active matter with pressure  $p_A$  from gauge fields) is:

$$T_{\mu\nu}^{(A)} = (\rho_A + p_A) u_\mu^{(A)} u_\nu^{(A)} + p_A g_{\mu\nu} \quad (13)$$

Phase B ghost quanta are pressureless ( $p_B = 0$ ):

$$T_{\mu\nu}^{(B)} = \rho_B u_\mu^{(B)} u_\nu^{(B)} \quad (14)$$

### 3.3 Gravitational Drag Interaction

The two-stream gravitational drag coupling is encoded in the minimal symmetric interaction tensor:

$$T_{\mu\nu}^{(\text{drag})} = -\beta \rho_A \rho_B (u_\mu^{(A)} u_\nu^{(B)} + u_\mu^{(B)} u_\nu^{(A)}) \quad (15)$$

where  $\beta \geq 0$  is the drag coupling constant (dimensions:  $[\text{energy density}]^{-1}$ ). The antisymmetric product is nonzero only in the  $rr$  sector, confirming that drag is purely radial.

### 3.4 Total EMT and Effective Fluid Parameters

The total energy-momentum tensor is:

$$T_{\mu\nu} = T_{\mu\nu}^{(A)} + T_{\mu\nu}^{(B)} + T_{\mu\nu}^{(\text{drag})} \quad (16)$$

The explicit components are:

$$T_{rr} = A[\rho_A + \rho_B + 2\beta\rho_A\rho_B] \quad (17)$$

$$T_{ij} = p_A a^2 \gamma_{ij} \quad (18)$$

Phase B contributes zero spatial pressure. Define effective fluid variables:

$$\rho_{\text{eff}} \equiv \rho_A + \rho_B + 2\beta\rho_A\rho_B, \quad p_{\text{eff}} \equiv p_A \quad (19)$$

### 3.5 Microphysical Origin of Phase B Quanta

The phenomenological description of Phase B as pressureless and non-interacting now follows from first principles via the Ginzburg–Landau analysis of Section 2.5.

When  $\langle\phi\rangle \rightarrow 0$  at  $r = R_{\text{max}}$ , the interaction Lagrangian:

$$\mathcal{L}_{\text{int}} = g \bar{\psi} \gamma^\mu A_\mu \psi + \dots \quad (20)$$

vanishes because  $A_\mu$  loses its coherent vacuum condensate. What remains are the free propagating momentum eigenstates of the formerly interacting fields—kinematic quanta without gauge dressing. Their properties follow directly:

**Pressurelessness.** Without interactions there is no mechanism to establish thermodynamic equilibrium. The quanta stream freely in momentum eigenstates, identical to pressureless cold matter:

$$p_B = \frac{1}{3} \langle v^2 \rangle \rho_B \rightarrow 0 \quad \left( \langle v^2 \rangle \rightarrow 0 \text{ as } \rho_A \rightarrow \rho_c \right) \quad (21)$$

**Non-interaction.** With no gauge condensate, all scattering cross sections vanish to all orders:

$$\sigma_{BB} \propto |\mathcal{M}|^2 \propto g^2 \langle \phi \rangle^2 = 0 \quad (22)$$

**Inward streaming.** At  $r = R_{\max}$  the shell has expanded to its maximum extent. Without radiation pressure from gauge interactions to resist it, the gravitational attraction of the central boundary at  $r = 0$  dominates and the quanta begin infalling.

**Decoherence radiation signature.** The Phase A  $\rightarrow$  Phase B transition radiates a brief burst of decoherence radiation, analogous to Unruh radiation at the transition surface. This produces a *stochastic gravitational wave burst* localised at  $r = R_{\max}$ , contributing to  $\Omega_{\text{GW}}^{\text{cycle}}(f)$  in equation (111) with characteristic frequency:

$$f_{\text{cycle}} = \frac{\mathcal{H}(R_{\max})}{2\pi} \approx \frac{H_0}{2\pi} \sqrt{\Omega_k} \left( \frac{R_{\max}}{a_0} \right)^{-1} \quad (23)$$

This is a third distinct testable stochastic GW component, in addition to the primordial and Phase B contributions identified in Section 12.

## 4 RCFM Field Equations

### 4.1 Einstein Field Equations

Applying  $G_{\mu\nu} = 8\pi G T_{\mu\nu}$  to the metric (1) with the total EMT (16), and using the Christoffel symbols:

$$\Gamma_{rr}^r = \frac{A'}{2A}, \quad \Gamma_{ij}^r = \frac{aa'}{A} \gamma_{ij}, \quad \Gamma_{rj}^i = \frac{a'}{a} \delta_j^i \quad (24)$$

we obtain two independent equations.

### 4.2 RCFM Friedmann Equation (F1)

The  $(r, r)$  Einstein equation gives the *constraint*:

$$\frac{(a')^2}{A} + 1 = \frac{8\pi G}{3} (\rho_A + \rho_B + 2\beta \rho_A \rho_B) a^2 \quad (\text{F1})$$

This is the RCFM Friedmann equation. In synchronous gauge ( $A = 1$ ) it reduces to:

$$(a')^2 = \frac{8\pi G}{3} \rho_{\text{eff}} a^2 - 1 \quad (25)$$

### 4.3 Raychaudhuri Equation (F2)

The spatial  $(i, j)$  Einstein equation gives the *dynamical* equation:

$$\frac{2}{\sqrt{A}} \frac{d\mathcal{H}}{dr} + 3\mathcal{H}^2 + \frac{1}{a^2} = -8\pi G p_A \quad (\text{F2})$$

Note the asymmetry: only  $p_A$  appears, since Phase B is pressureless.

## 4.4 Conservation Equations

The Bianchi identity  $\nabla_\mu T^{\mu\nu} = 0$  yields two coupled conservation equations with the constant drag rate  $\Gamma_{\text{drag}}$  (derived in Section 10):

$$\rho'_A + 3\mathcal{H}\sqrt{A}(\rho_A + p_A) = +\Gamma_{\text{drag}} \rho_A \rho_B \quad (\text{C1})$$

$$\rho'_B - 3\mathcal{H}\sqrt{A}\rho_B = -\Gamma_{\text{drag}} \rho_A \rho_B \quad (\text{C2})$$

The signs encode the physical exchange: Phase B loses energy to Phase A via gravitational drag. The total  $\nabla_\mu T^{\mu\nu} = 0$  is satisfied by construction.

## 4.5 Effective Gravitational Constant

The Phase B density evolves as:

$$\bar{\rho}_B(r) = \rho_{B,0} \left( \frac{R_{\text{max}}}{a(r)} \right)^3 \quad (26)$$

The RCFM effective gravitational constant is:

$$G_{\text{eff}}(r) = G \left[ 1 + 2\beta \rho_{B,0} \left( \frac{R_{\text{max}}}{a(r)} \right)^3 \right] \quad (27)$$

Current observations constrain the dimensionless correction:

$$\Lambda_{\text{RCFM}}(r) \equiv 2\beta \rho_{B,0} \left( \frac{R_{\text{max}}}{a(r)} \right)^3 \ll 1 \quad (28)$$

## 4.6 Recovery of $\Lambda$ CDM as a Limiting Case

When  $\beta \rightarrow 0$  and  $\rho_B \rightarrow \Lambda/(8\pi G) = \text{const}$ , Eq. (F1) reduces to the standard Friedmann equation with a cosmological constant. The RCFM therefore contains  $\Lambda$ CDM as a limiting case.

# 5 Phase A Equation of State

## 5.1 Relativistic Thermal Fluid

Phase A is Standard Model matter in thermal equilibrium. For a relativistic gas at temperature  $T$ :

$$\rho_A = \frac{\pi^2}{30} g_*(T) T^4, \quad p_A = \frac{\rho_A}{3} \cdot \frac{g_{*P}(T)}{g_*(T)} \quad (29)$$

where  $g_*(T)$  counts the relativistic energy degrees of freedom and  $g_{*P}(T)$  the pressure degrees of freedom.

## 5.2 Multi-Epoch Structure

The equation of state parameter  $w_A$  evolves through several epochs as the shell expands and cools:

Table 1: Phase A equation of state across cosmological epochs.

| Epoch          | $T$ range                      | $g_*$                    | $w_A$       |
|----------------|--------------------------------|--------------------------|-------------|
| Electroweak    | $T \gg 100 \text{ GeV}$        | 106.75                   | 1/3         |
| QCD transition | $T \sim 150 \text{ MeV}$       | $10.75 \rightarrow 3.36$ | $\sim 1/3$  |
| Lepton era     | $1 \text{ MeV} \lesssim T$     | 10.75                    | 1/3         |
| BBN            | $T \sim 0.1 \text{ MeV}$       | 3.36                     | 1/3         |
| Matter dom.    | $T \ll \text{eV}$              | —                        | $\approx 0$ |
| Dilution       | $r \rightarrow R_{\text{max}}$ | —                        | ill-defined |

## 5.3 Unified Analytic Parameterisation

A closed-form expression valid across all epochs is:

$$w_A(\rho_A) = \frac{1}{3} \cdot \frac{1}{1 + (\rho_{\text{eq}}/\rho_A)^{1/4}} \quad (30)$$

where  $\rho_{\text{eq}} \approx 4.15 \times 10^{-14} \text{ J m}^{-3}$  is the density at matter-radiation equality. This gives  $w_A \rightarrow 1/3$  for  $\rho_A \gg \rho_{\text{eq}}$  and  $w_A \rightarrow 0$  for  $\rho_A \ll \rho_{\text{eq}}$ .

## 5.4 Mapping $r$ to Redshift

The temperature-scale factor relation from photon number conservation is:

$$T(r) = T_0 \cdot \frac{a_0}{a(r)}, \quad \rho_A(r) = \rho_{A,0} \cdot \frac{g_*(T(r))}{g_*(T_0)} \cdot \left( \frac{a_0}{a(r)} \right)^4 \quad (31)$$

This provides the complete dictionary between the RCFM radial coordinate  $r$  and standard cosmological observables.

# 6 Weak-Field Limit and Newtonian Gravity

## 6.1 Metric Perturbations

We perturb the background around  $(\bar{a}, \bar{\rho}_A, \bar{\rho}_B, \bar{p}_A)$  satisfying (F1)–(F2). In longitudinal gauge:

$$ds^2 = -(1 + 2\Phi) dr^2 + \bar{a}^2(1 - 2\Phi) d\Omega_3^2 \quad (32)$$

where  $|\Phi| \ll 1$  is the Bardeen gravitational potential [3].

## 6.2 Poisson Equation

In the sub-horizon, quasi-static limit  $k^2/\bar{a}^2 \gg \bar{\mathcal{H}}^2$ , the perturbed  $(r, r)$  Einstein equation reduces to:

$$\nabla^2 \Phi = 4\pi G_{\text{eff}} \delta\rho_A \quad (33)$$

This is the Newtonian Poisson equation with effective gravitational constant  $G_{\text{eff}}$  given by (27). Newtonian gravity is fully recovered.

## 6.3 Geodesic Equation

In the non-relativistic, sub-Hubble limit, the spatial components of the Phase A geodesic equation give:

$$\ddot{\mathbf{x}} = -\nabla\Phi \quad (34)$$

confirming Newton's second law with gravitational potential sourced by (33).

## 6.4 Observational Constraint on $G_{\text{eff}}$

Lunar laser ranging and pulsar timing constrain time-variation of  $G$  [8]:

$$\frac{G_{\text{eff}} - G}{G} < 10^{-5} \quad (35)$$

This gives the upper bound on the drag coupling:

$$\beta < \frac{10^{-5}}{2\rho_{B,0}} \left( \frac{a_0}{R_{\text{max}}} \right)^3 \quad (36)$$

# 7 CMB Angular Power Spectrum

## 7.1 Mode Functions on $S^3$

Scalar perturbations are expanded in hyperspherical harmonics  $Q_{\ell m}^{(n)}$  satisfying:

$$\nabla_{S^3}^2 Q_{\ell m}^{(n)} = -\frac{n^2 - 1}{a^2} Q_{\ell m}^{(n)}, \quad n = 1, 2, 3, \dots, \quad \ell \leq n - 1 \quad (37)$$

The effective comoving wavenumber  $k_n^2 = (n^2 - 1)/a^2$  is discrete, in contrast to the continuous spectrum in flat space. Discreteness is observationally relevant only for  $\ell \lesssim 10$ .

## 7.2 Conformal Coordinate

Define the conformal radial coordinate:

$$\eta(r) \equiv \int_0^r \frac{dr'}{a(r')} \quad (38)$$

so that  $d\eta = dr/a$ . All perturbation equations are written with dots denoting  $d/d\eta$ .

### 7.3 Boltzmann Hierarchy

The photon temperature multipole moments  $\Theta_\ell(k, \eta)$  satisfy the Boltzmann hierarchy [18]:

$$\dot{\Theta}_0 = -k\Theta_1 - \dot{\Phi} \quad (39)$$

$$\dot{\Theta}_1 = \frac{k}{3}\Theta_0 - \frac{2k}{3}\Theta_2 + \frac{k}{3}\Phi + \dot{\kappa}\left(\frac{v_b}{3} - \Theta_1\right) \quad (40)$$

$$\dot{\Theta}_\ell = \frac{k}{2\ell+1}[\ell\Theta_{\ell-1} - (\ell+1)\Theta_{\ell+1}] + \dot{\kappa}\left[-\Theta_\ell + \delta_{\ell 0}\Theta_0 + \delta_{\ell 1}\frac{v_b}{3} - \delta_{\ell 2}\frac{\Pi}{10}\right], \quad \ell \geq 2 \quad (41)$$

where  $\Pi = \Theta_2 + \Theta_{P,2} + \Theta_{P,0}$ ,  $v_b$  is the baryon velocity, and  $\dot{\kappa} = an_e\sigma_T$ .

The gravitational potential is sourced by:

$$k^2\Phi = 4\pi G_{\text{eff}}(\eta) a^2[\delta\rho_A + \delta\rho_B(1 + 2\beta\bar{\rho}_A)] \quad (42)$$

### 7.4 Tight-Coupling Approximation and Sound Speed

Before recombination ( $\dot{\kappa} \rightarrow \infty$ ):

$$c_s^2(\eta) = \frac{1}{3(1 + R_b(\eta))}, \quad R_b(\eta) \equiv \frac{3\rho_A^{(b)}}{4\rho_\gamma} \quad (43)$$

### 7.5 RCFM Oscillator Equation

In tight coupling, combining (39) and (40) yields:

$$\ddot{\Theta}_0 + \frac{\dot{R}_b}{1 + R_b}\dot{\Theta}_0 + k^2 c_s^2 \Theta_0 = \mathcal{F}_{\text{grav}}(\eta) + \mathcal{S}_B(\eta) \quad (44)$$

The standard gravitational driving term is:

$$\mathcal{F}_{\text{grav}} = -k^2 \left[ \frac{\Phi}{3} + \frac{\dot{R}_b \Phi}{k^2(1 + R_b)} \right] - \ddot{\Phi} \quad (45)$$

The Phase B source term (absent in  $\Lambda$ CDM) is:

$$\mathcal{S}_B(\eta) = -\frac{4\pi G \bar{\rho}_A a^2 \cdot 2\beta \bar{\rho}_B \delta_B}{1 + R_b} \quad (46)$$

### 7.6 Sound Horizon

$$r_s(\eta_*) = \int_0^{\eta_*} c_s(\eta') d\eta' \approx \frac{2}{\sqrt{3}} \frac{1}{\sqrt{\Omega_b h^2 / \Omega_\gamma h^2}} \ln \left[ \frac{\sqrt{1 + R_b(\eta_*)} + \sqrt{R_b(\eta_*) + R_{b,\text{eq}}}}{1 + \sqrt{R_{b,\text{eq}}}} \right] \quad (47)$$

The RCFM correction to the sound horizon is  $\Delta r_s / r_s \sim \Lambda_{\text{CMB}} \ll 1$ .

## 7.7 Silk Damping

$$r_d^2(\eta) = \int_0^\eta \frac{d\eta'}{6(1+R_b)\dot{\kappa}} \left[ \frac{16}{15} + \frac{R_b^2}{1+R_b} \right] \quad (48)$$

The RCFM modifies this through the BBN helium abundance. The enhanced  $G_{\text{eff}}$  at BBN shifts  $Y_p$ :

$$\Delta Y_p \approx 0.16 \Lambda_{\text{BBN}} = 0.16 \times 2\beta \rho_{B,0} \left( \frac{R_{\text{max}}}{a_{\text{BBN}}} \right)^3 \quad (49)$$

## 7.8 Transfer Function and Line-of-Sight Integration

Using the Seljak–Zaldarriaga line-of-sight method [18]:

$$\Delta_\ell(k) = \int_0^{\eta_0} S(k, \eta) j_\ell[k(\eta_0 - \eta)] d\eta \quad (50)$$

$$\Delta_\ell(k) = \Delta_\ell^{(0)}(k) + \Delta_\ell^{(B)}(k) + \Delta_\ell^{\text{ISW}}(k) \quad (51)$$

## 7.9 Angular Power Spectrum

$$C_\ell^{\text{RCFM}} = \frac{2}{\pi} \int_0^\infty k^2 \left| \Delta_\ell^{(0)}(k) + \Delta_\ell^{(B)}(k) + \Delta_\ell^{\text{ISW}}(k) \right|^2 P_\Phi(k) dk \quad (52)$$

## 7.10 Acoustic Peak Structure

Peak positions occur at  $kr_s(\eta_*) = n\pi$ :

$$\ell_n \approx n\pi \frac{\Delta\eta_*}{r_s(\eta_*)} \quad (53)$$

$$\ell_n^{\text{RCFM}} > \ell_n^{\Lambda\text{CDM}} \quad (54)$$

The Phase B source modifies the odd/even peak amplitude ratio:

$$\mathcal{R}_n \equiv \frac{C_{\ell_{2n-1}}}{C_{\ell_{2n}}} \quad (55)$$

$$\mathcal{R}_n^{\text{RCFM}} = \mathcal{R}_n^{\Lambda\text{CDM}} [1 + \epsilon_n(\beta, \rho_{B,0}, R_{\text{max}})] \quad (56)$$

where  $\epsilon_n$  grows with  $n$ , providing a scale-dependent signature testable with Planck, ACT, and SPT [13].

## 7.11 Low- $\ell$ Quadrupole Suppression

$$\frac{C_2^{S^3}}{C_2^{\mathbb{R}^3}} = \left( \frac{k_{\text{min}} \Delta\eta_*}{2\pi} \right)^{n_s-1} < 1 \quad (57)$$

providing a natural explanation for the observed low- $\ell$  CMB power deficit.

## 7.12 Late Integrated Sachs-Wolfe Effect

$$\dot{\Phi} = -\frac{\dot{G}_{\text{eff}}}{G_{\text{eff}}}\Phi + \text{standard GR terms} \quad (58)$$

At  $a \rightarrow a_0$ ,  $\dot{G}_{\text{eff}} \rightarrow 0$ , so the late ISW approaches the  $\Lambda$ CDM result. At intermediate redshifts  $1 < z < 10$ , a measurable excess ISW signal relative to  $\Lambda$ CDM is predicted.

## 8 Matter Power Spectrum

### 8.1 Growth Factor

$$\delta_A'' + 2\mathcal{H}\delta_A' - 4\pi G_{\text{eff}}\bar{\rho}_A a^2 \delta_A = \mathcal{S}(k, r) \quad (59)$$

where  $\mathcal{S} = 4\pi G a^2 \bar{\rho}_A \cdot 2\beta \bar{\rho}_B \delta_B$ .

$$D_+(r) = \mathcal{H}(r) \int_0^r \frac{dr'}{\mathcal{H}^3(r') a^2(r')} \quad (60)$$

$$f\sigma_8(r) = \sigma_8 \Omega_A^\gamma(r) \cdot \frac{D_+(r)}{D_+(r_0)} \quad (61)$$

$$\gamma_{\text{RCFM}} = \gamma_{\text{GR}} + \frac{\Lambda_{\text{RCFM}}(r)}{1 + \Lambda_{\text{RCFM}}(r)} \Delta\gamma \quad (62)$$

### 8.2 Matter Transfer Function

$$T_A(k) = T_A^{\text{BBKS}}(k) \cdot \mathcal{M}(k) \quad (63)$$

$$T_A^{\text{BBKS}}(q) = \frac{\ln(1 + 2.34q)}{2.34q} \left[ 1 + 3.89q + (16.1q)^2 + (5.46q)^3 + (6.71q)^4 \right]^{-1/4} \quad (64)$$

$$\mathcal{M}(k) = 1 + \alpha_\beta \ln\left(\frac{k}{k_{\text{eq}}}\right) \cdot \Lambda_{\text{eq}} \quad (65)$$

### 8.3 Full Power Spectrum

$$P(k, r_0) = P_{\text{ini}}(k) \cdot T_A^2(k) \cdot D_+^2(r_0) \cdot \mathcal{M}^2(k) + P^{(B)}(k) \quad (66)$$

$$P^{(B)}(k) = \left( \frac{D_+(r_0)}{D_+(r_*)} \right)^2 \int_0^{r_*} \int_0^{r_*} G_+(r_0, r_1) G_+(r_0, r_2) \mathcal{S}(k, r_1) \mathcal{S}^*(k, r_2) dr_1 dr_2 \quad (67)$$

### 8.4 BAO Scale

$$k_n^{\text{BAO}} = \frac{n\pi}{r_s(\eta_*)} - n\pi \frac{\Delta r_s}{r_s^2(\eta_*)} \quad (68)$$

Internal consistency requires that BAO and CMB peak shifts be equal in magnitude—a *joint-fit consistency check* for the model.

## 9 Primordial Spectrum from the Singularity

### 9.1 The Singularity as a Quantum Boundary

Near  $r = 0$ , with  $a(r) \propto r^\nu$ , the mode equation:

$$u_k'' + (k^2 - V_{\text{eff}}(r)) u_k = 0 \quad (69)$$

has effective potential:

$$V_{\text{eff}}(r) = \frac{\mu^2 - \frac{1}{4}}{r^2}, \quad \mu^2 = \left(\nu - \frac{1}{2}\right)^2 + \ell(\ell + 2) \quad (70)$$

The exact solutions are Bessel functions:

$$u_k(r) = \sqrt{r} [C_1 J_\mu(kr) + C_2 Y_\mu(kr)] \quad (71)$$

Regularity at the origin requires  $C_2 = 0$ .

### 9.2 Scale Invariance Condition

In the long-wavelength limit  $kr \ll 1$ :

$$|u_k|^2 \propto k^{-2\mu} \implies \mathcal{P}_\Phi(k) \propto k^{3-2\mu} \quad (72)$$

Exact scale invariance ( $n_s = 1$ ) requires  $\mu = 3/2$ , which for  $\ell = 0$  gives:

$$\nu = 2 \quad (73)$$

### 9.3 Self-Consistency with (F1)

Substituting  $a(r) = a_1 r^\nu$  into the RCFM Friedmann equation near  $r = 0$ , where Phase B dominates:

$$\frac{\nu^2}{r^2} = \frac{8\pi G \rho_{B,0} R_{\text{max}}^3}{3a_1} \cdot r^{-\nu} \quad (74)$$

Matching powers:  $-2 = -\nu$ , giving:

$$\nu = 2 \quad (\text{self-consistent}) \quad (75)$$

### 9.4 Spectral Tilt

Including the next-order correction:

$$a(r) = a_1 r^2 (1 + \epsilon_1 r + \dots) \quad (76)$$

the spectral tilt is:

$$n_s - 1 = -\epsilon_1 / k_* \quad (77)$$

The Planck 2018 measurement  $n_s = 0.9649 \pm 0.0042$  [13] constrains:

$$\epsilon_1 = (1 - n_s) \cdot k_* \approx 0.035 k_*, \quad k_* = 0.05 \text{ Mpc}^{-1} \quad (78)$$

## 9.5 First-Principles Derivation of $\epsilon_1$

In the original formulation,  $\epsilon_1$  was treated as a free expansion coefficient requiring numerical calibration. We now derive it analytically from equation (F1).

Substituting  $a(r) = a_1 r^2 (1 + \epsilon_1 r)$  into (F1) in synchronous gauge and expanding in powers of  $r$ , with Phase B dominating near the singularity ( $\rho_B \propto r^{-6} a_1^{-3}$ ), the coefficient at order  $r^{-1}$  yields:

$$4a_1\epsilon_1 = \frac{8\pi G}{3} \cdot \frac{3\rho_{B,0} R_{\max}^3 \epsilon_1}{a_1^2} \quad (79)$$

Solving for  $\epsilon_1$ :

$$\epsilon_1 = \sqrt{\frac{2\pi G \rho_{B,0} R_{\max}^3}{a_1^3}} \quad (80)$$

This is not a free parameter:  $\epsilon_1$  is determined entirely by  $G$ ,  $\rho_{B,0}$ ,  $R_{\max}$ , and  $a_1$ . Substituting into (77), the spectral tilt becomes:

$$n_s - 1 = -\frac{1}{k_*} \sqrt{\frac{2\pi G \rho_{B,0} R_{\max}^3}{a_1^3}} \quad (81)$$

Matching to the Planck value  $n_s = 0.9649$  gives a *joint constraint*:

$$\frac{\rho_{B,0} R_{\max}^3}{a_1^3} = \frac{(1 - n_s)^2 k_*^2}{2\pi G} \approx \frac{(0.035)^2 (0.05 \text{ Mpc}^{-1})^2}{2\pi G} \quad (82)$$

This is directly testable through independent measurements of  $\rho_{B,0}$ ,  $R_{\max}$ , and  $a_1$  via the observables listed in Table 2.

**Result 1.** The spectral tilt parameter  $\epsilon_1$  is fully determined by the RCFM field equations and the Planck-measured value of  $n_s$ . The model contains no undetermined spectral parameters.

## 9.6 Tensor Perturbations and Consistency Relation

$$n_T = -2\epsilon_1/k_* = -(1 - n_s) \quad (83)$$

This reproduces the single-field inflation consistency relation  $n_T = -r_T/8$  from a completely different physical mechanism.

# 10 Drag Kernel and Microphysics

## 10.1 Gravitational Scattering Cross Section

At leading order in  $G$ , the momentum transfer rate per unit volume between the two streams is:

$$\dot{p}_{\text{drag}}(r) = \frac{4\pi G^2 \rho_A \rho_B}{v_{\text{rel}}^3(r)} \quad (84)$$

Comparing with (15):

$$\beta(r) = \frac{4\pi G^2}{v_{\text{rel}}^3(r) \cdot c^2} \quad (85)$$

## 10.2 Relative Velocity Profile

Near  $r = 0$ , with  $a \propto r^2$ :

$$v_{\text{rel}}(r) \approx \frac{\mathcal{C}}{r}, \quad \mathcal{C} \equiv 2 + \sqrt{\frac{8\pi G \rho_{B,0} R_{\text{max}}^3}{3a_1}} \quad (86)$$

## 10.3 Constant Drag Rate

$$\beta(r) \cdot \mathcal{D}(r) = \frac{4\pi G^2}{\mathcal{C}^2 a_1 c^2} \equiv \Gamma_{\text{drag}} = \text{const} \quad (87)$$

The  $r$ -dependences cancel exactly, a consequence of the conformal symmetry imposed by  $a \propto r^2$ .

**Result 2.** The drag rate  $\Gamma_{\text{drag}}$  is a true constant of the model, determined entirely by  $G$ ,  $c$ , and  $a_1$ . The continuity equations (C1)–(C2) are fully closed.

## 10.4 Fixed-Point Analysis

Setting  $\rho'_A = \rho'_B = 0$  in (C1)–(C2) reveals two fixed points. Linearising around the matter-dominated fixed point ( $\rho_A \rightarrow \rho_c$ ,  $\rho_B \rightarrow 0$ ), both eigenvalues of the Jacobian have negative real parts when  $\mathcal{H} > 0$  and  $w_A > -1$ . The matter-dominated state is a *stable attractor*—the cycle is self-sustaining.

## 10.5 Perturbative Stability of $\Gamma_{\text{drag}}$

We now demonstrate that the constancy of  $\Gamma_{\text{drag}}$  is robust against perturbations of the background solution.

Let  $a(r) = a_1 r^2 (1 + \delta(r))$  with  $|\delta| \ll 1$ . The perturbed Hubble rate is:

$$\mathcal{H}(r) = \frac{2}{r} \left( 1 + \frac{r\delta'}{2} + O(\delta^2) \right) \quad (88)$$

The perturbed velocity profiles are:

$$v_{\text{rel}}(r) = \frac{\mathcal{C}}{r} (1 + O(\delta')) \quad (89)$$

The perturbed drag coupling and kernel are:

$$\beta(r) = \frac{4\pi G^2 r^3}{\mathcal{C}^3 c^2} (1 + O(\delta')) \quad (90)$$

$$\mathcal{D}(r) = \frac{\mathcal{C}}{a_1 r^3} (1 + O(\delta)) \quad (91)$$

Their product:

$$\Gamma_{\text{drag}} = \beta(r) \cdot \mathcal{D}(r) = \frac{4\pi G^2}{\mathcal{C}^2 a_1 c^2} (1 + O(\delta, \delta')) \quad (92)$$

The first-order corrections cancel identically. The cancellation is a consequence of the conformal symmetry at the singularity: any perturbation preserving the  $r^2$  UV scaling of  $a(r)$  leaves  $\Gamma_{\text{drag}}$  invariant. This can be stated precisely as:

$$\left. \frac{d\Gamma_{\text{drag}}}{d \ln r} \right|_{r \rightarrow 0} = 0 \quad (\text{exact under conformal scaling}) \quad (93)$$

Second-order corrections give:

$$\frac{\delta\Gamma_{\text{drag}}}{\Gamma_{\text{drag}}} = O\left(\frac{\epsilon_1^2}{k_*^2}\right) \approx 1.2 \times 10^{-3} \quad (94)$$

which is observationally negligible.

**Result 3.** The constancy of  $\Gamma_{\text{drag}}$  is perturbatively stable to first order in deviations from the exact  $a \propto r^2$  background. Second-order corrections are of order  $\epsilon_1^2/k_*^2 \sim 10^{-3}$ .

## 11 Entropy and the Arrow of Time

### 11.1 Bogoliubov Transformation at the Singularity

The reprocessing operator at  $r = 0$  relates incoming Phase B modes  $\hat{\chi}_{\text{in}}(k)$  to outgoing Phase A modes  $\hat{\phi}_{\text{out}}(k)$  via a Bogoliubov–Valatin transformation:

$$\hat{\phi}_{\text{out}}(k) = \alpha(k)\hat{\chi}_{\text{in}}(k) + \beta_k\hat{\chi}_{\text{in}}^\dagger(-k) + \hat{\xi}(k) \quad (95)$$

Unitarity requires  $|\alpha(k)|^2 - |\beta_k|^2 = 1$ . The vacuum noise operator  $\hat{\xi}(k)$  carries only amplitude information, not phase structure.

### 11.2 Entropy of the Outgoing State and Global Conservation

The von Neumann entropy of the Phase A output state is:

$$S_A^{\text{out}} = \sum_k \left[ (1 + |\beta_k|^2) \ln(1 + |\beta_k|^2) - |\beta_k|^2 \ln |\beta_k|^2 \right] \quad (96)$$

From (85),  $|\beta_k|^2 \propto v_{\text{rel}}^{-3} \rightarrow 0$  as  $r \rightarrow 0$ . Therefore:

$$S_A^{\text{out}} \Big|_{r=0} = 0 \quad (97)$$

At first sight this appears to violate the generalised second law. The resolution requires tracking the *total* von Neumann entropy of the composite system:

$$S_{\text{total}} = S_A + S_B + S_{AB}^{(\text{ent})} \quad (98)$$

where  $S_{AB}^{(\text{ent})}$  is the bipartite entanglement entropy between the two streams.

During the infall phase ( $R_{\text{max}} > r > 0$ ), gravitational drag generates growing entanglement between the streams:

$$\frac{dS_{AB}^{(\text{ent})}}{dr} = -\Gamma_{\text{drag}} \frac{\rho_A \rho_B}{\rho_{\text{total}}} \cdot \mathcal{F}(v_{\text{rel}}) \quad (99)$$

As  $S_A$  decreases,  $S_{AB}^{(\text{ent})}$  increases by the same amount: entropy is transferred from local Phase A entropy into bipartite entanglement, not destroyed.

At  $r \rightarrow 0$ , the reduced density matrix of Phase A alone has zero entropy—but only because all entropy has been moved into the entanglement term:

$$S_A^{\text{out}}|_{r=0} = 0, \quad S_{AB}^{(\text{ent})}|_{r=0} = S_{\text{total}} = \text{const} \quad (100)$$

This is the exact analogue of *Page's theorem* in black hole evaporation [11]: the entropy of the Hawking radiation subsystem alone goes to zero when the internal and external degrees of freedom are maximally entangled, while the total entropy is conserved. The generalised second law is satisfied globally.

**Result 4.** Thermodynamic entropy is genuinely reset to zero at each passage through the singularity. The thermodynamic arrow of time is re-established at each cycle. This resolves the cyclic entropy paradox without information loss—the incoming Phase B momentum information is diluted into the vacuum noise operator  $\hat{\xi}$ , seeding only the amplitude  $\mathcal{A}_s$  of the primordial power spectrum. Global entropy is conserved via entanglement transfer.

### 11.3 Information Conservation and Quantum Scrambling

The scattering matrix at the singularity boundary is:

$$\mathbf{S} = \begin{pmatrix} \alpha(k) & \beta_k^* \\ \beta_k & \alpha^*(k) \end{pmatrix}, \quad \mathbf{S}\mathbf{S}^\dagger = \mathbf{1} \quad (101)$$

Unitarity ( $|\alpha|^2 - |\beta_k|^2 = 1$ ) guarantees that the map from incoming Phase B states to outgoing Phase A states is invertible. Information is conserved.

The vacuum noise operator satisfies:

$$\langle \hat{\xi}(k) \hat{\xi}^\dagger(k') \rangle = \mathcal{A}_s(k) \delta^{(3)}(k - k') \quad (102)$$

The phase information of  $\hat{\chi}_{\text{in}}(k)$  is encoded in the relative phases of  $\alpha(k)$  and  $\beta_k$ , which become unobservable to Phase A observers through decoherence into environmental modes. This is quantum *scrambling*, not information loss, analogous to black hole complementarity [16]. The amplitude  $\mathcal{A}_s$  of the primordial power spectrum therefore encodes a fingerprint of the previous cycle.

## 12 Gravitational Wave Propagation

### 12.1 Tensor Perturbations on $S^3$

$$\nabla_{S^3}^2 \mathcal{G}_{ij}^{(n)} = -(n^2 - 3) \mathcal{G}_{ij}^{(n)}, \quad n \geq 2 \quad (103)$$

The minimum mode is  $n = 2$ ; there are no  $n = 0, 1$  tensor modes on  $S^3$ .

### 12.2 Gravitational Wave Equation

$$h_n'' + 3\mathcal{H}h_n' + \left[ \frac{n^2 - 3}{a^2} + m_{\text{GW}}^2(r) \right] h_n = \mathcal{T}_n^{(B)}(r) \quad (104)$$

$$m_{\text{GW}}^2(r) = \frac{8\pi G_{\text{eff}}(r)}{a^2} \Gamma_{\text{drag}}(\rho_A - \rho_B) \quad (105)$$

### 12.3 Phase B Anisotropic Stress Source

$$\pi_{ij}^{(B)} = \rho_B v_B^2 \left( \hat{r}_i \hat{r}_j - \frac{1}{3} \gamma_{ij} \right), \quad \mathcal{T}_n^{(B)} = \frac{16\pi G_{\text{eff}} \rho_B v_B^2}{3} \delta_{n,2} \quad (106)$$

### 12.4 Gravitational Wave Speed

$$c_T^2(r) \approx 1 - \alpha_T(r), \quad \alpha_T(r) = \left. \frac{d \ln G_{\text{eff}}}{d \ln a} \right|_r = \frac{-6\Lambda_{\text{RCFM}}(r)}{1 + \Lambda_{\text{RCFM}}(r)} \quad (107)$$

### 12.5 GW170817 Constraint

[10]:

$$\left| \frac{c_T - c}{c} \right| < 5 \times 10^{-16} \quad (108)$$

Applied to (107):

$$\Lambda_{\text{RSD}}(z=0) = 2\beta_0 \rho_{B,0} \left( \frac{R_{\text{max}}}{a_0} \right)^3 < 1.7 \times 10^{-16} \quad (109)$$

### 12.6 Low-Frequency Cutoff

$$f_{\text{cut}} = \frac{c m_{\text{GW}}(r_0)}{2\pi\hbar} = \frac{c}{2\pi\hbar} \sqrt{\frac{8\pi G_{\text{eff}} \Gamma_{\text{drag}} |\rho_A - \rho_B|}{a_0^2}} \quad (110)$$

### 12.7 Stochastic Background Components

$$\Omega_{\text{GW}}(f) = \Omega_{\text{GW}}^{\text{prim}}(f) + \Omega_{\text{GW}}^{(B)}(f) + \Omega_{\text{GW}}^{\text{cycle}}(f) \quad (111)$$

The three components are:

- (i) **Primordial** ( $\Omega_{\text{GW}}^{\text{prim}}$ ): Power law with  $n_T = -(1 - n_s) \approx -0.035$ , consistent with LIGO O3 upper limits.
- (ii) **Phase B source** ( $\Omega_{\text{GW}}^{(B)}$ ): Monochromatic at  $f^{(B)} \approx 3.5 \times 10^{-19} \text{ Hz}$  (sub-astrometric band).
- (iii) **Decoherence burst** ( $\Omega_{\text{GW}}^{\text{cycle}}$ ): Stochastic burst at  $f_{\text{cycle}}$  from the Phase A→B transition at  $r = R_{\text{max}}$  (Section 3.5, eq. (23)). Potentially observable with LISA and pulsar timing arrays.

## 13 Observational Predictions

### 13.1 Parameter Degeneracy Breaking

The three free RCFM parameters  $(\beta, \rho_{B,0}, R_{\max})$  appear in the single observable combination:

$$\Lambda_{\text{RCFM}}(z) = 2\beta\rho_{B,0} \left( \frac{R_{\max}}{a(z)} \right)^3 \quad (112)$$

Note that  $R_{\max}$  is now constrained by (10) and  $\epsilon_1$  is fixed by (80), reducing the effective free parameter count. Four observational handles probe  $\Lambda_{\text{RCFM}}$  at different redshifts:

Table 2: Observational constraints on  $\Lambda_{\text{RCFM}}$  and their redshift sensitivity.

| Observable         | Dataset          | Redshift                    | Sensitivity            |
|--------------------|------------------|-----------------------------|------------------------|
| CMB peak positions | Planck 2018 [13] | $z_* \approx 1100$          | $\Lambda_{\text{CMB}}$ |
| BAO scale          | DESI DR1 [5]     | $z \approx 0.3\text{--}2.1$ | $\Lambda_{\text{BAO}}$ |
| BBN helium         | Cooke et al. [4] | $z \approx 6 \times 10^8$   | $\Lambda_{\text{BBN}}$ |
| $f\sigma_8$ (RSD)  | BOSS+eBOSS [1]   | $z \approx 0.5$             | $\Lambda_{\text{RSD}}$ |
| GW speed           | GW170817 [10]    | $z \approx 0.009$           | $\Lambda_{\text{RSD}}$ |

### 13.2 Internal Consistency Test

Since  $\ln \Lambda_{\text{RCFM}}(z) = \ln(2\beta\rho_{B,0}R_{\max}^3) - 3\ln a(z)$ , all four  $\Lambda$  values must lie on a straight line of slope  $-3$  in the  $\ln \Lambda$ – $\ln a$  plane. Deviation from slope  $-3$  falsifies the model.

### 13.3 Naturalness of the GW170817 Bound

The bound  $\Lambda_{\text{RSD}} < 1.7 \times 10^{-16}$  might appear to require fine-tuning. We now show it is naturally satisfied.

The physical content of (109) is:

$$2\beta_0\rho_{B,0} \left( \frac{R_{\max}}{a_0} \right)^3 = 2\beta_0\rho_{\Lambda} < 1.7 \times 10^{-16} \quad (113)$$

where we have used the observational identification  $8\pi G\rho_{B,0}(R_{\max}/a_0)^3/3 \approx H_0^2\Omega_{\Lambda}$ , so  $\rho_{B,0}(R_{\max}/a_0)^3 \sim \rho_{\Lambda}$ .

The drag coupling  $\beta$  has dimensions  $[\text{energy density}]^{-1}$ . Its natural scale is set by the Planck density:

$$\beta_0^{\text{natural}} \sim \rho_{\text{Pl}}^{-1} = \left( \frac{c^5}{\hbar G^2} \right)^{-1} \quad (114)$$

Therefore:

$$\Lambda_{\text{RSD}}^{\text{natural}} \sim \frac{2\rho_{\Lambda}}{\rho_{\text{Pl}}} \sim 2 \times \frac{6 \times 10^{-27} \text{ kg m}^{-3}}{10^{96} \text{ kg m}^{-3}} \sim 10^{-122} \quad (115)$$

This is *orders of magnitude below* the GW170817 constraint  $1.7 \times 10^{-16}$ . The model therefore has enormous headroom:  $\beta_0$  can be  $10^{106}$  times larger than its Planck-suppressed natural value before violating the observational bound.

**Result 5.** No fine-tuning is required to satisfy the GW170817 constraint. For any  $\beta_0$  between the Planck-suppressed value and  $10^{106} \times \rho_{\text{Pl}}^{-1}$ , the model is consistent with all gravitational wave observations.

## 13.4 Summary of RCFM vs. $\Lambda$ CDM Predictions

Table 3: Key observational predictions of RCFM compared to  $\Lambda$ CDM.

| Observable                     | $\Lambda$ CDM           | RCFM                                        | Testable?        |
|--------------------------------|-------------------------|---------------------------------------------|------------------|
| Acceleration                   | Cosmological $\Lambda$  | Ghost quanta drag,<br>$w_B \approx -1$      | ✓                |
| CMB peak $\ell_n$              | Standard                | Shifted to higher $\ell$                    | ✓                |
| Odd/even ratio $\mathcal{R}_n$ | Fixed by $\Omega_b$     | Scale-dependent $\epsilon_n$                | ✓                |
| Quadrupole $C_2$               | Predicted; observed low | Naturally suppressed ( $S^3$ cutoff)        | Qualitative      |
| $G_{\text{eff}}$ variation     | None                    | $\propto (1+z)^3$                           | ✓ (BBN, pulsar)  |
| $f\sigma_8$                    | $\gamma \approx 0.55$   | $\gamma_{\text{RCFM}} > \gamma_{\text{GR}}$ | ✓ (Euclid)       |
| $n_T$                          | $-r_T/8$                | $-(1-n_s)$                                  | ✓ (CMB-S4)       |
| GW speed                       | $c_T = c$               | $c_T = c(1 + 3\Lambda_{\text{RSD}})$        | ✓ (GW170817)     |
| Low-freq. GW cutoff            | Absent                  | At $f_{\text{cut}}$                         | ✓ (LISA)         |
| Decoherence                    | Absent                  | At $f_{\text{cycle}}$ , eq. (23)            | ✓ (LISA, PTA)    |
| GW burst                       |                         |                                             |                  |
| Entropy reset                  | Not addressed           | Genuine ( $S = 0$ at $r = 0$ )              | —                |
| $\rho_c$ value                 | Free parameter          | Fixed to $\rho_{\text{EW}}$                 | ✓ (EW precision) |
| $\epsilon_1$ value             | Free                    | Eq. (80)                                    | ✓ (Planck)       |

## 14 Numerical Framework

### 14.1 Background Integration

The complete RCFM background ODE system in synchronous gauge ( $A = 1$ ) is:

$$a' = a\mathcal{H} \quad (116)$$

$$\mathcal{H}^2 = \frac{8\pi G}{3}(\rho_A + \rho_B + 2\beta\rho_A\rho_B)a^2 - 1 \quad (117)$$

$$\rho'_A = -3\mathcal{H}(\rho_A + p_A) + \Gamma_{\text{drag}}\rho_A\rho_B \quad (118)$$

$$\rho'_B = +3\mathcal{H}\rho_B - \Gamma_{\text{drag}}\rho_A\rho_B \quad (119)$$

with boundary conditions at  $r = r_\epsilon \ll 1$ :

$$a(r_\epsilon) = a_1 r_\epsilon^2, \quad \rho_A(r_\epsilon) = 0, \quad \rho_B(r_\epsilon) = \rho_{B,0}(R_{\text{max}}/a(r_\epsilon))^3 \quad (120)$$

## 14.2 Boltzmann Code Architecture

The perturbation solver extends a standard flat- $\Lambda$ CDM integrator with three RCFM-specific modifications:

- (i) Background quantities ( $\mathcal{H}$ ,  $G_{\text{eff}}$ ,  $\rho_B$ ) interpolated from the modified background solutions (116)–(119).
- (ii) Phase B source terms (46), (67), (106) added to the Boltzmann hierarchy.
- (iii) Mode functions computed on  $S^3$  using discrete wavenumbers  $k_n = \sqrt{n^2 - 1}/a_0$ .

## 14.3 Joint Likelihood

The effective free parameters after applying the constraints of Sections 2.5 and 9.5 are  $(\beta, \rho_{B,0})$ , with  $R_{\text{max}}$  fixed by (10) and  $\Gamma_{\text{drag}}$  by (87). They are determined by joint minimisation of:

$$\chi_{\text{total}}^2 = \chi_{\text{CMB}}^2 + \chi_{P(k)}^2 + \chi_{\text{BBN}}^2 + \chi_{f\sigma_8}^2 \quad (121)$$

subject to the joint constraint (82).

**Note:** Numerical integration of (116)–(119) and full Boltzmann solver implementation are reserved for a companion paper.

## 15 Conclusion

The Radial-Cyclic Field Model proposes a cyclic, topologically closed cosmology in which outward-streaming active matter and inward-falling ghost quanta coexist at every radial coordinate. Starting from the metric ansatz (1) and the dual-stream energy-momentum tensor (16), we have derived a complete, closed system of modified Friedmann and perturbation equations.

In this version of the paper, seven previously open theoretical problems have been resolved:

1. **Phase transition origin.** The critical density  $\rho_c$  is derived from a Ginzburg–Landau effective potential for the gauge-sector VEV, fixing  $\rho_c = \rho_{\text{EW}}$  from known electroweak physics and removing it as a free parameter.
2. **Ghost quanta microphysics.** Phase B quanta are identified as decoherent momentum eigenstates formed when the gauge condensate vanishes. Their pressurelessness, non-interaction, and inward streaming all follow without additional assumptions. A new stochastic GW burst signature at frequency  $f_{\text{cycle}}$  is predicted.
3. **Robustness of  $\Gamma_{\text{drag}}$ .** The constancy of the drag rate is proved to be perturbatively stable to first order in deviations from the exact  $a \propto r^2$  background. Second-order corrections are  $O(10^{-3})$ .
4. **Thermodynamic consistency.** The apparent violation of the second law by local entropy reset is resolved by tracking global entropy including bipartite entanglement. Total entropy is conserved in exact analogy with Page’s theorem.

5. **Naturalness.** The GW170817 constraint  $\Lambda_{\text{RSD}} < 1.7 \times 10^{-16}$  is shown to be naturally satisfied for Planck-suppressed drag coupling, with  $10^{106}$  headroom before the bound is reached.
6. **First-principles spectral tilt.** The parameter  $\epsilon_1$  is derived analytically from the RCFM Friedmann equation, yielding the joint constraint (82) directly testable against Planck data.
7. **Information conservation.** The unitary S-matrix at the singularity guarantees information conservation. Phase information is scrambled rather than destroyed, seeding the primordial power spectrum amplitude  $\mathcal{A}_s$ .

The principal structural results remain:

1. The RCFM Friedmann equation (F1) contains  $\Lambda$ CDM as a limiting case.
2. Newtonian gravity is exactly recovered in the sub-horizon, non-relativistic limit.
3. The singularity boundary condition  $a \propto r^2$  is self-consistent with the field equations and uniquely generates a nearly scale-invariant primordial spectrum, making inflation unnecessary.
4. The drag rate  $\Gamma_{\text{drag}}$  is exactly constant, closing the system analytically.
5. Thermodynamic entropy is genuinely reset to zero at each cycle via a Bogoliubov transformation.
6. Gravitational waves propagate at speed  $c_T \approx c$  to order  $\Lambda_{\text{RSD}}$ , consistent with the GW170817 constraint.
7. The model makes at least seven observationally distinct predictions relative to  $\Lambda$ CDM, testable with existing and near-future datasets.

The remaining work before quantitative comparison with data is the numerical integration of the background equations and Boltzmann solver implementation, which will be presented in a companion paper.

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[To be completed.]

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